Daoguan NING

Postdoc Researcher

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Employment

Stanford University Nov. 01, 2024 – Present
Postdoc Researcher
Advisor: Prof. Xiaolin Zheng
Research: Fast recycling of rare-earth materials | Synthesis of metal-based energetic materials | Optical measurement of energetic-material combustion

Technical University of Darmstadt, Germany Oct. 15, 2022 – Sep. 30, 2024
Postdoc Researcher
Advisor: Prof. Andreas Dreizler
High-speed optical diagnostics of aluminum-steam combustion | Multi-parameter optical measurement of iron particle ignition and combustion | Theoretical analysis of heterogeneous combustion of metal particles

Eindhoven University of Technology, Netherlands Jul. 01, 2022 – Sept. 30, 2022
Postdoc Researcher (with Ph.D. advisor immediately following degree conferral)
Advisor: Prof. Philip de Goey
Experimental study of lifted iron flames | Optical measurement of iron particle combustion | Development of novel iron combustion strategy

Education

Eindhoven University of Technology, Netherlands Sept. 1, 2018 – Jun. 30, 2022
Doctor of Philosophy, Mechanical Engineering
Advisor: Prof. Philip de Goey
The date of Ph.D. degree conferral: Jun. 30, 2022

Huazhong University of Science and Technology, China Sept. 2015 – Jun. 2018
Master of Engineering Science, Engineering Thermophysics
GPA: 8.6/10 (thesis 9.3/10)

Jiangsu University of Science and Technology, China Sept. 2011 – Jun. 2015
Bachelor of Engineering, Thermal Energy and Power Engineering
GPA: 4.2/5

Competencies

Expertise	Metal combustion | Optical thermometry | High-speed imaging | Emission spectroscopy | Theory of metal particle combustion
Lab experience	Design of experiments | Construction of optical diagnostic setup | Operation of high-power laser
Programming	MATLAB | FORTRAN | Python
Professional software	Cantera | Chemkin | SolidWorks | Ansys Fluent

Publications

†These authors contribute equally to the paper. * Corresponding author.

- **2026**

30. M. Szedlock, D. Ka, Y. Li, M. Kiefer, D. Kong, D. Ning, W. Kau, W. Tarpeh, X. Zheng*. *'Self-Propagating Aluminothermic Reduction for Recycling Key Battery Materials'*. , (In Process).
29. D. Ka, D. Ning, S.D. Tse*, X. Zheng*. *'Stretch-controlled flame propagation of solid fuel/oxidizer composites'*. Proceedings of the Combustion Institute 41 (Under Review).
28. A. Huynha, J. Leem, D. Ning, J. Kalman, X. Zheng*. *'Dual-Function Graphene Oxide Aerogels for Enhanced Mechanical and Combustion Properties of Boron-Based Solid Propellants'*. Proceedings of the Combustion Institute 41 (Under Review).
27. Y. Li, T. Li*, L. Choisez, C. Ruan, L. Ahmad, D. Ning, A. Dreizler. *'Ignition temperature of porous hydrogen direct reduced iron particles with inhomogeneous impurities'*. Proceedings of the Combustion Institute 41 (Under Review).
26. D. Ning*. *'Steam-diluted iron combustion enabling simultaneous burn-rate enhancement and nanoparticle mitigation'*. International Journal of Heat and Mass Transfer 264, 128730.
25. S. Shi, D. Ning*, T. Krenn, H. Chen, C. Ruan, T. Li, A. Dreizle. *'Combustion characteristics of submillimeter aluminum droplets in high-temperature steam'*. Chemical Engineering Journal (Under Review).
24. D. Ning, D. Ka, A.H. Huynh, Y. Li, X. Zheng*. *'Ignition Temperature and Combustion Dynamics of B-HTPB Composite Microparticles'*. Combustion and Flame 286, 114804.
23. J. Chen, K. Li, F. Peng, D. Ning, X.C. Mi, Y. Zheng, S. Xu*, D. Chen, X. Wen, Y. Liu, W. Cai*. *'Combustion characteristics of single iron particles under ammonia co-firing conditions'*. Combustion and Flame 285, 114780.

- **2025**

22. T. Li, B. Nguyen, Y. Gao, L. Elsässer, D. Ning, A. Scholtissek, A. van Duin, C. Hasse, B. Böhm. *'Critical nanoparticle formation in iron combustion: single particle experiments with in-situ multi-parameter diagnostics aided by multi-scale simulations'*. Fuel 404, 136303.
21. D. Ning*, A. Scholtissek, A. Dreizler. *'A novel approach for modeling iron microparticle oxidation during the reactive cooling process'*. Combustion and Flame 279, 114313.
20. B. Nguyen*, A. Scholtissek, T. Li, D. Ning, O. Stein, A. Dreizler, C. Hasse. *'Nanoparticle formation in the boundary layer of burning iron microparticles: Modeling and simulation'*. Chemical Engineering Journal 507 (2025), 160039.
19. A. Sperling, M. P. Deutschmann, D. Ning*, J. Spielmann, T. Li, U. I. Kramm, H. Nirschl, B. Böhm, A. Dreizler. *'Oxidation progress and inner structure during single micron-sized iron particles combustion in a hot gas atmosphere'*. Fuel 381, 133147.

- **2024**

18. D. Ning*, Y. Li, T. Li, B Böhm, A. Dreizler. *'Size-resolved ignition temperatures of isolated iron microparticles'*. Combustion and Flame 270, 113779.
17. D. Ning*, T. Li, B. Böhm, A. Dreizler. *'Temperature of micrometric burning iron particles with in-situ resolved initial sizes'*. Combustion and Flame 270, 113737.
16. D. Ning*, A. Dreizler. *'A quantitative theory for heterogeneous combustion of nonvolatile metal particles in the diffusion-limited regime'*. Combustion and Flame 269, 113692.
15. L. Thijs†, D. Ning*†, Y. Shoshin, T. Hazenberg, X.C. Mi, J. A. van Oijen, P. de Goey *'Temperature evolution of laser-ignited micrometric iron particles: A comprehensive experimental dataset and numerical assessment of laser heating impact'*. Applications in Energy and Combustion Science 19, 100284.
14. J. Mich*, A. K. da Silva, D. Ning, T. Li, D. Raabe, B. Böhm, A. Dreizler, C. Hasse, A. Scholtissek *'Modeling the oxidation of iron microparticles during the reactive cooling phase'*. Proceedings of the Combustion Institute 40 (1-4), 105538.
13. B. Nguyen*, D. Braig, A. Scholtissek, D. Ning, T. Li, D. Raabe, A. Dreizler, C. Hasse. *'Ignition and kinetic-limited oxidation analysis of single iron microparticles in a hot laminar flow'*. Fuel 371, 131866.

- **2023**

12. D. Ning*, T. Hazenberg, Y. Shoshin, J.A. van Oijen, G. Finotello, L.P.H. de Goey. *'Experimental and theoretical study of single iron particle combustion under low-oxygen dilution conditions'*. Fuel 357, 129718. (ESI Highly Cited Paper)
11. D. Ning*, T. Li, J. Mich, A. Scholtissek, B. Böhm, A. Dreizler. *'Multi-stage oxidation of iron particles in a flame-generated hot laminar flow'*. Combustion and Flame 256, 112950.
10. D. Ning*, Y. Shoshin. *'Thermal inertia effect of reactive sources on one-dimensional discrete combustion wave propagation'*. Combustion and Flame 253, 112790.

- **2022**

9. D. Ning*, Y. Shoshin, J. van Oijen, G. Finotello, L.P.H. de Goey. *'Size evolution during laser-ignited single iron particle combustion'*. Proceedings of the Combustion Institute 39 (3), 3561-3571.
8. D. Ning*, Y. Shoshin, J. van Oijen, G Finotello, L.P.H. de Goey. *'Critical temperature for nanoparticle cloud formation during combustion of single micron-sized iron particle'*. Combustion and Flame 244, 112296.
7. D. Ning*, Y. Shoshin, M. van Stiphout, J. van Oijen, G. Finotello, L.P.H. de Goey. *'Temperature and phase transitions of laser-ignited single iron particle'*. Combustion and Flame 236, 111801. (Top 8 Cited Paper in Combust. Flame over 06/2020–06/2025)

- **2021**

6. D. Ning*, Y. Shoshin, J. van Oijen, G. Finotello, L.P.H. de Goey. *'Burn time and combustion regime of laser-ignited single iron particle'*. Combustion and Flame 230, 111424

- **2018**

5. D. Ning, S. Wang, A. Fan*, H. Yao. *'A numerical study of the effects of CO_2 and H_2O on the ignition characteristics of syngas in a micro flow reactor'*. International Journal of Hydrogen energy 43 (50), 22649-22657

- **2017**

4. D. Ning, S. Wang, A. Fan*, H. Yao. *'Effect of radiation emission and reabsorption on flame temperature and NO formation in H_2/CO/air counterflow diffusion flames'*. International Journal of Hydrogen Energy 42 (34), 22015-22026.
3. D. Ning, Y. Liu, Y. Xiang, A. Fan*. *'Experimental investigation on non-premixed methane/air combustion in Y-shaped meso-scale combustors with/without fibrous porous media'*. Energy Conversion and Management 138, 22-29.
2. D. Ning, Y. Liu, Y. Xiang, A. Fan*. *'Effects of fuel composition and strain rate on NO emission of premixed counter-flow H_2/CO/air flames'*. Energy Conversion and Management 123, 402-409.

- **2016**

1. Y. Liu, D. Ning, A. Fan*, H. Yao. *'Experimental and numerical investigations on flame stability of methane/air mixtures in mesoscale combustors filled with fibrous porous media'*. International Journal of Hydrogen Energy 42 (34), 22015-22026.

Teaching and Supervision

- **Teaching experience**
  - *'Lecture on fundamental theories and diagnostic techniques of metal fuel combustion'*. Jan. 2023 – Aug. 2024
  - Invited lecture: *'Fundamentals of Carbon-free Iron Combustion: Insights from Single Particle Experiments'*. Xinghuo Forum, Shanghai Jiao Tong University, Shanghai, China (20 Oct. 2023)
- **Master thesis**
  - Yuan L. Zhang. *'Theoretical analysis of micron-sized aluminium particle combustion in water vapor'*. Nov. 2023 – Present
  - Yu H. Li. *'Simultaneous high-speed optical measurements of ignition temperature of iron particles in a novel drop-tube reactor'*. Jun. 2023 – Dec. 2023

- Xiang Li. *'Design and commissioning of a novel drop-tube reactor for studying iron particle ignition and combustion'*. Nov. 2022 – Jul. 2023
- Koen M.J. van Duijnhoven. *'Towards systematic characterization of lifted micro iron flames'*. Mar. 2022 – Sept. 2022
- Jeroen J.A.H. Peeters. *'Combustion of micron sized iron particles'*. Feb. 2019 – Aug. 2019

- **Bachelor Thesis**
  - Lemar Abdi. *'Reduced-order discrete source model for iron particles'*. Feb. 2022 – Jul. 2022
- **Intership project**
  - Rodrigo Solipa. *'Chemical kinetic estimation for the reactive quenching process of a single iron particle'*. Sept. 2021 – Jan. 2022

Achievements

- DoD DURIP Award (2025). (PI: Xiaolin Zheng; I contributed as a primary proposal writer.)
- Energy and Environment Seed Fund (€40.000) awarded by Technical University of Darmstadt (2024)
- Heroes for Heroes Grant (€7.500) awarded by Eindhoven University of Technology (2022)
- National Postgraduate Scholarship from Ministry of Education of the People's Republic of China (2017)
- Merit Undergraduate Student (2014)